

Charles Nguyen

Computer Science B.S. @ UC San Diego

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Summary

I am a fourth-year Computer Science undergraduate at the University of California, San Diego, with a strong focus on front-end engineering and scalable full-stack development. I have hands-on experience in building and modernizing production-ready systems with an emphasis on security, maintainability, and user-centered design.

As Vice Chair of Finance, Vice Chair of Projects, and Webmaster for the IEEE Student Branch at UC San Diego, I lead both technical initiatives and organizational operations. I spearheaded the modernization of IEEE's technical infrastructure by architecting and deploying a containerized backend system with version control integration, OpenID authentication, and Docker-based deployment pipelines. I am also developing a chatbot that leverages codebase indexing through retrieval-augmented generation (RAG) to provide up-to-date and context-aware responses.

As a Lab Fellow for HDSI Lab 3.0, I actively introduce AI solutions across diverse projects, including image analysis and voice-to-speech interpretation, to enhance research and instructional workflows. In addition, I am currently developing an expense tracking application that uses AI-based receipt parsing to extract transaction details and streamline bill splitting among friends.

I am also an Okta Certified Professional, demonstrating expertise in identity and access management, authentication protocols, and secure enterprise-grade authentication systems.

Experience

IEEE at UC San Diego	San Diego, CA
Webmaster	October 2023 - Present
As Webmaster, I architect and maintain IEEE UCSD's technical infrastructure, spanning dashboards, authentication systems, internal tooling, and self-hosted production services.	

I built the IEEE Dashboard using Convex as the backend and TanStack Start as the frontend, creating a centralized platform for event discovery, reimbursement submissions, event management, sponsorship tracking, resume database access, link sharing, user management, and analytics. I integrated Google APIs, including Google Calendar synchronization, to ensure real-time event updates and improved member engagement across the organization.

I led the initiative to self-host IEEE's core infrastructure by deploying our dashboard, main website, authentication services, and internal tools on a VPS environment. Using open-source tooling such as Dokploy, I manage containerized services including databases, authentication providers, and internal microservices. This increased reliability, security, and operational control while reducing reliance on external platforms.

I implemented organization-wide authentication using Logto with OpenID Connect and JWT-based security, establishing role-based access control across all IEEE services. This unified authentication layer secures internal dashboards, project tools, and administrative systems under a standardized identity infrastructure.

Beyond web systems, I developed the IEEE Presence Tracker using Convex as the backend and a Raspberry Pi with AstroJS as the frontend. The system leverages Rust-based Bluetooth proximity detection to automatically determine member attendance in real time. I also built the IEEE Parts Platform, featuring a 2D blueprint interface for inventory visualization and a structured check-in and check-out system powered by Convex and TanStack Start.

<https://ieeeatucsd.org>

HDSI Lab 3.0

Lab Fellow

Halıcıoğlu Data Science Institute

November 2025 - Present

As a Lab 3.0 Fellow at HDSI, I contribute to the lab's technical infrastructure, student-facing platforms, and outreach initiatives supporting data science and engineering education.

I developed the lab's public-facing website using AstroJS, building a structured platform to showcase projects, events, and initiatives through Markdown-driven content. I implemented the table of contents, drafted the Privacy Policy, and ensured the site complied with COPPA standards. The platform includes dynamic member listings and is fully responsive across desktop and mobile devices.

I also built a collaborative code-sharing and debugging platform using AstroJS and Convex to support partnerships between the lab and local high schools. The platform allows students to share and debug code online without requiring account creation. It supports versioning, structured log visualization, syntax highlighting, and diff comparison, providing an accessible and lightweight development environment for educational use.

Building on an existing open-source inventory tracker, I am extending the system to support a graphical user interface and potential mobile integration to enable student clock-in and clock-out tracking. This improves operational visibility and simplifies administrative workflows.

In addition to core development work, I provide technical guidance across lab projects, including UI design support, AI prompt engineering, and integration of tools such as OpenRouter. I occasionally participate in outreach efforts, assisting with school engagement activities and providing debugging support for the Robot Flower project.

Through full-stack development, systems thinking, and educational outreach, I help scale Lab 3.0's technical capabilities while supporting broader community engagement.

<https://lab3.ucsd.edu>

Projects

Personal Website

September 2023 - Present

Designed and developed a personal portfolio website using **AstroJS** to showcase technical skills, projects, and professional background. Integrated a blog section for publishing articles on programming and other topics of interest. Drew inspiration from a range of modern design styles to create a clean, responsive, and user-friendly interface. Through this project, gained in-depth experience with Astro's component architecture, routing, and content management features, significantly strengthening frontend development skills.

<https://chark1es.dev>

Share This Site

September 2025 - now

A lightweight web tool that enables users to generate temporary, shareable links that automatically expire after a specified duration, ranging from 15 minutes to 7 days. Prioritizing speed, simplicity, and privacy, this tool eliminates the need for user login and automatically secures URLs with HTTPS. Its design aims to facilitate fast and secure short-term sharing while minimizing user data collection.

<https://sharethis.site>

Who can work

September 2025 - now

Developed a lightweight scheduling web app for groups to swiftly discover shared availability. The app features an interactive time-grid user interface, shareable links, and a privacy-first, no-account design. It prioritizes simplicity, performance, and accessibility across various devices.

<https://whocan.work>

Moozy

July 2025 - Present

My role in this app is to turn UI designs into functional components for a mobile app. This mobile app uses flutter and firebase and my role is to make sure the components are properly working, handle any potential bugs, and prepare the app for a production-ready scale.

IEEE at UCSD Dashboard

October 2023 - Present

The dashboard is a centralized way for users, officers, and sponsors to view their stats, submit applications, and download forms. This project was built using AstroJS + TypeScript as the frontend and Firebase + RestAPI as the backend. I was the first to design and implement all functionality within the dashboard, including the events, reimbursement, and attendance system.

<https://ieeeeatucsd.org/dashboard>

ChopChop

Jan 2026 - current

Full-stack expense-splitting mobile app built with **Expo Router + React Native** (frontend) and **Convex** (backend). Implemented Google OAuth onboarding, real-time friend-based balance tracking, receipt scanning with on-device OCR + LLM parsing (cloud fallback via OpenRouter), flexible split logic (even/percent/custom/itemized), and settlement-backed ledger architecture ensuring accurate “you owe / you’re owed” calculations.

Volunteering

GDPT Anoma

June 2010 - Present

Gardena, CA

As a Team Lead in Anoma, I work as a role model for younger students to help them learn more about the Vietnamese culture as well with assisting the teachers and other leaders in making food, charity donations, and hosting cultural events.

Skills

Docker	AstroJS	Markdown & MDX	Tailwind CSS	OpenID Connect
NoSQL	SMTP	JS & TS	Python	HTML

Awards

Best General Officer

Jun 2025

IEEE at UC San Diego

Best General Officer

Jun 2024

IEEE at UC San Diego

Certifications

Microsoft Office Specialist: Word Associate (Office 2019)

February 2021

Microsoft

<https://www.credly.com/badges/ef0a1a75-459d-421a-9c70-9c6eba8ff01f/email>

Okta Certified Professional

January 2025

Okta

https://www.credly.com/badges/92844853-f3bc-49e7-8e46-b7de59722cb5/public_url

References

Saura Naderi	K-14 Outreach Director	Erik Duarte	2025-26 IEEE UC San Diego President	Rachel Levy	Supervisor at Chadwick School
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